Problem_Set2_code_2

October 20, 2026

0.1 Question 3(e)

[13]:
```
import numpy as np
import pandas as pd
from scipy import stats
```

[14]:
```
wage = pd.read_excel('wage.xlsx')
wage['logwage'] = np.log(wage['wage'])
wage.head()
```

[14]:

	male	wage	logwage
0	0	6.315296	1.842975
1	0	5.479770	1.701063
2	0	3.642170	1.292580
3	0	4.593337	1.524607
4	0	2.418158	0.883006

[22]:
```
wage_males = wage[wage['male']==1]
wage_females = wage[wage['male']==0]

n1 = len(wage_females)
n2 = len(wage_males)

y1 = wage_females['logwage']
y2 = wage_males['logwage']
y1_bar = np.mean(y1)
y2_bar = np.mean(y2)

s1 = np.sum((y1-y1_bar)**2) / (n1-1)
s2 = np.sum((y2-y2_bar)**2) / (n2-1)

T = np.sqrt(n1*n2/ (n1+n2)) * (np.log(s1) - np.log(s2))

k_hat = (n1+n2)* (np.sum((y1-y1_bar)**4) + np.sum((y2-y2_bar)**4)) / (np.
 ↪sum((y1-y1_bar)**2) + np.sum((y2-y2_bar)**2)) **2

z = T / np.sqrt(k_hat-1)
p_robust = 1- stats.norm.cdf(z)
```

```
F = s1/s2
p_normal = 1 - stats.f.cdf(F,n1-1,n2-1)

print(f'p-value robust to non-normality {p_robust}')
print(f'p value under normality {p_normal}')
```

```
p-value robust to non-normality 0.17481673039385381
p value under normality 0.04734658854081053
```

The p-value under normal assumptions shows that we would reject the null with 0.05 significance level, while the non-normality robust test showa we would not reject the null.